

YASH MAHESH BANGERA

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years of experience architecting and deploying production-grade Machine Learning, Generative AI, and MLOps solutions on AWS for Fortune 500 enterprises across healthcare, finance, IT, and manufacturing. Proven record designing agentic AI systems, RAG pipelines, and LLM-powered automation using Amazon Bedrock, SageMaker, LangChain, and LangGraph — driving up to 80% efficiency gains and a 60% reduction in post-deployment issues. AWS-certified (Machine Learning Engineer, AI Practitioner, Solutions Architect) with deep expertise translating ambiguous enterprise requirements into scalable, secure, customer-facing AI solutions. Actively seeking AI Engineer, Machine Learning Engineer, Data Scientist, and Forward Deployed Engineer opportunities in the UAE and Singapore.

EDUCATION

M.S., Data Science — Rochester Institute of Technology

Rochester, NY | May 2022

B.Tech, Computer Science & Engineering — Vellore Institute of Technology

Chennai, India | May 2019

TECHNICAL SKILLS

- Generative AI & LLM Engineering:** Amazon Bedrock, Bedrock AgentCore, LangChain, LangGraph, Strands Agents, LlamaIndex, HuggingFace Transformers, RAG Architectures, vLLM, LangSmith, RAGAS, Vector DBs (FAISS, Pinecone, pgvector, ChromaDB, Qdrant)
- Machine Learning & Deep Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Computer Vision (OpenCV), NLP (NLTK, SpaCy, Gensim)
- MLOps & Cloud Infrastructure:** Amazon SageMaker, MLflow, Airflow, Kubeflow, Docker, Kubernetes, AWS CDK, Terraform, GitHub Actions, GitLab CI/CD
- AWS Services:** SageMaker, Bedrock, Textract, Rekognition, Comprehend, Personalize, A2I, API Gateway, Lambda, S3, Glue, Athena, DynamoDB, CloudFront, IAM, KMS, VPC, SNS, SQS, CloudWatch
- Programming & Data:** Python, R, Java, C/C++, MATLAB, SQL, Pandas, NumPy, Jupyter, Tableau
- Full-Stack & Dev Tools:** React.js, TypeScript, Streamlit, Git, VS Code, PyCharm

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- AWS Certified AI Practitioner
- AWS Certified Solutions Architect – Associate
- Machine Learning – Stanford University (Coursera)

PROFESSIONAL EXPERIENCE

Associate Delivery Consultant, AI/ML — Amazon Web Services (AWS)

Dec 2025 – June 2026

- Partner directly with enterprise customers' product, engineering, and data science teams to translate ambiguous requirements into production-grade AI/ML solutions using SageMaker and Bedrock, unlocking 20–30% potential revenue optimization — operating as a forward-deployed technical advisor embedded in customer environments.
- Architected an intelligent document processing pipeline (Amazon Bedrock, Lambda, S3, SQS, OpenSearch, OCR, NLP) enabling workforce re-allocation, cutting costs up to 40%, and delivering 70–80% efficiency gains.
- Unified classical ML, GenAI, and LLM Ops practices with AWS AI/ML services; automated deployment via AWS CDK/Terraform and CI/CD, reducing deployment time 50% while strengthening security posture.

Cloud Support Engineer I, SageMaker, Vision & Other Services — Amazon Web Services (AWS)

Jun 2022 – Dec 2025

- Owned resolution of 500+ critical, escalated SageMaker issues for Fortune 500 customers across healthcare, finance, IT, and manufacturing — delivering real-time inference, latency optimization, and custom model hosting solutions that cut post-deployment issues 60%.
- Built internal GenAI- and ML-powered automation tools using prompt engineering, cutting case resolution time across a 20+ engineer team and improving customer engagement quality.
- Ranked #2 globally within Amazon SageMaker Support Operations for case complexity and resolution speed; sustained 95%+ CSAT across 7+ consecutive quarters and co-authored standard operating procedures adopted team-wide.

Teaching & Research Assistant — Rochester Institute of Technology

Mar 2021 – Dec 2021

- Mentored 30+ students in Data Science Fundamentals and Python, driving 90% proficiency in building and deploying ML models (K-NN, Neural Networks, Decision Trees) through targeted feedback achieving 95%+ accuracy.
- Engineered NLP feature pipelines (Pandas, NLTK) across 500K+ data points for two linguistic research studies, producing 10 new features (word embeddings, GloVe, TF-IDF) and cutting model training time 35% while lifting accuracy to 85%+.

Lead Machine Learning Engineer — Omdena, Inc.

Oct 2019 – Jul 2020

- Led a computer vision team using ResNet to detect Amazon Rainforest wildfires, cutting false positives 10–33% via Gaussian blur removal and reaching 95–97% model accuracy.
- Engineered 10 statistical features and built a DBSCAN clustering model (scikit-learn) powering an ethical credit-scoring system for underbanked Indonesian borrowers.
- Analyzed COVID-19 policy impacts on vulnerable populations (Tableau, SQL, NLTK); findings presented at the UN AI for Good Global Summit and showcased by NASDAQ in Times Square.

Research & Development Intern — AI Square Global Solutions LLP

May 2019 – Jul 2020

- Built a healthcare-sector NLP dataset from 10,000+ scraped news articles and applied 7 HuggingFace summarization models (evaluated via ROUGE/BLEU), lifting downstream analysis accuracy 15% and customer satisfaction 50%.

KEY PROJECTS

Intelligent Document Processing Pipeline — AWS CDK, Textract, Comprehend, Bedrock

Jan – Mar 2026

- Cut infrastructure provisioning time 60% and operational costs 32% by architecting a fully automated CI/CD pipeline (GitLab, SQS, SNS) that eliminated manual deployments end-to-end.
- Engineered a serverless OCR/NLP pipeline (Python, AWS CDK, Textract, Comprehend, Bedrock) extracting structured insights — summaries, entities, keywords — from thousands of documents across 10 formats.
- Achieved 100% fault-tolerant processing by parallelizing the workflow with SQS and Lambda, including automatic retry and partial batch-failure handling.
- Secured the pipeline end-to-end using S3 pre-signed URLs, protecting data in transit and at rest.

RAG-Based Contextual Chatbot — Amazon Bedrock & OpenSearch

Oct – Dec 2024

- Boosted query resolution rates 80% by building a Retrieval-Augmented Generation chatbot on Amazon Bedrock Knowledge Bases powered by Claude models.
- Eliminated 100% of manual ingestion effort by orchestrating automated S3-to-Bedrock pipelines for 1,000+ product documents.
- Cut manual document look-up time 70% through semantic indexing (cosine-similarity embeddings) in Amazon OpenSearch Serverless.
- Elevated response relevance and production-readiness using metadata filtering, chunking optimization, re-ranking, and LLM evaluation metrics (context relevance, faithfulness).

Language Variety / Dialect Identification for Similar Languages — English, Portuguese, Spanish — NLP

Jan – Dec 2021

- Delivered a capstone project distinguishing similar language varieties across three languages — English, Portuguese, and Spanish — using natural language processing.
- Curated the first-ever gold-annotated dataset (DSL-TL) for the task via crowdsourcing on Amazon MTurk.
- Built a neural network model to distinguish child languages of a parent language with 95% accuracy.
- Co-authored a research paper aggregating the project's findings, published at LREC-COLING 2024.

Forest Fire Detection via Computer Vision — Amazon Rainforest

Oct – Dec 2020

- Conducted data curation of ground-truth labels for nearly 9,000 images using LabelBox, cutting raw image curation effort 100%.
- Integrated images into a data augmentation and cleaning pipeline for Gaussian noise removal (Jupyter, Pandas, NumPy, OpenCV), producing a standard dataset for model training.
- Trained ResNet-50 computer vision models for image segmentation, achieving 95–97% accuracy on unseen data and optimizing detection time 80% upon deployment.

PUBLICATIONS

- Language Variety Identification with True Labels — LREC-COLING 2024
- Privacy Preservation and Signature Aggregation for Medical Data — ICRTAC 2019